Data ScienceTech Institute (DSTI)

Applied MSc in Data Science and Artificial Intelligence

Deep Learning Project Report

Skin Cancer Classification

Presented by

TRAORE Sy Lucien

Hanna Hellgren

Sanae Dariouche

Academic year : 2024-2025

Table of Contents

LIST OF TABLES

LIST OF FIGURES

Abstract

Skin cancer is among the most common types of cancer worldwide, and early detection is crucial for successful treatment. This project aims to develop a deep learning model capable of classifying different types of skin lesions using images from the HAM10000 dataset. Leveraging EfficientNet-B0, a state-of-the-art convolutional neural network architecture, we seek to achieve high accuracy while maintaining model efficiency suitable for real-world deployment.

We replaced ResNet-50 with EfficientNet-B0—dramatically reducing parameter count (25.6M $\rightarrow$ 5.3M) while boosting ImageNet top-1 accuracy from 74.9% to 77.1%. Key training strategies included advanced data augmentations, weighted sampling to counter class imbalance, mixed-precision acceleration for 2–3$\times$ speedup on modern GPUs with PyTorch, and a `ReduceLROnPlateau` scheduler. Our final model achieved 86% accuracy, 86.5% precision, 86.0% recall, and 86.2% F1—a robust, balanced performance across seven classes.

Introduction

Image classification is one of the fundamental tasks in computer vision, with real-life applications ranging from medical imaging and environmental monitoring to vehicle automation. It allows us to categorize images by feeding them into a model which then outputs a prediction for the class an image could belong to. Image classification works by training a model on an extensive dataset, from which the model learns the features and patterns of each class of images.

There are several different types of image classification, depending on the complexity of the task and the kind of images used. The most common are binary classification, which involves categorizing images into two classes, and multiclass classification, which involves more than two classes. To associate one image with multiple classes, one would use multilabel classification. And if those classes have different levels, hierarchical classification can be used.

Pre-trained models

Pre-trained models are neural networks that are trained on large and general-purpose datasets, before being tuned for specific tasks. Thanks to the extensive pre-training, these models can perform well with limited data while saving time and computational resources.

The HAM10000 dataset comprises seven lesion categories with a heavy imbalance (e.g., AKIEC $\approx$ 1341 pictures vs. DF $\approx$ 23) using PyTorch. Traditional CNNs like ResNet-50 ($\approx$ 25.6M parameters) struggle under tight resource budgets and class imbalance. We innovated by adopting EfficientNet-B0, which uses Mobile Inverted Bottleneck layers and compound scaling to jointly balance depth, width, and resolution—delivering a 77.1% top-1 ImageNet accuracy with just 5.3M parameters.

0.1 Dataset

The HAM10000 dataset includes 10,015 dermatoscopic images classified into :

- NV : Melanocytic nevi
- MEL : Melanoma
- BKL : Benign keratosis-like lesions
- BCC : Basal cell carcinoma
- AKIEC : Actinic keratoses
- VASC : Vascular lesions
- DF : Dermatofibroma

An 80%-20% train-validation split was used with class stratification.

0.2 Model Innovation : ResNet vs. EfficientNet-B0

- **Parameter Efficiency** : ResNet-50 (25.6M) vs. EfficientNet-B0 (5.3M).
- **Accuracy Boost** : EfficientNet-B0 achieves 77.1% top-1 ImageNet accuracy vs. 74.9% for ResNet-50.
- **Compound Scaling** : Simultaneous scaling of depth, width, and resolution.
- **Advanced Blocks** : MBConv and Squeeze-and-Excitation improve representational capacity.

0.3 Training Methodology

0.3.1 Data Handling & Augmentation

- **Augmentations** : Random resized crops, flips, rotations, color jitter, Gaussian blur — improve generalization on subtle lesion features.
- **Weighted Sampling** : We used PyTorch's `WeightedRandomSampler` to oversample minority classes proportionally, ensuring balanced batches.

0.3.2 Optimization

- **Loss** : `CrossEntropyLoss` combines log-softmax and `NLLLoss`, ideal for multi-class tasks (PyTorch).
- **Optimizer** : Adam with weight decay (1×10^{-4}) for stable convergence.
- **Scheduler** : `ReduceLROnPlateau` reduces learning rate on validation loss plateau, preventing wasted epochs.
- **Mixed Precision** : Used `torch.cuda.amp.autocast` and `GradScaler` for 2–3× speed improvements on Tesla/RTX GPUs (PyTorch).

0.4 Metrics Explained

- **Accuracy** : Ratio of correct predictions.
- **Precision** : $\frac{TP}{TP+FP}$
- **Recall (Sensitivity)** : $\frac{TP}{TP+FN}$
- **F1 Score** : Harmonic mean of precision and recall : $2 \times \frac{Precision \times Recall}{Precision + Recall}$.
- **Confusion Matrix** : Summarizes TP, FP, TN, FN per class.

0.5 Results Interpretation

Epoch 30/30

- Train ▷ loss = 0.0781, accuracy = 97.43%
- Validation ▷ loss = 0.4747, accuracy = 86.02%
- ▷ New Best Saved Model

Metric and Confusion Matrix

Evaluation of the Final Model :

- Accuracy : 86.02%
- Precision : 86.50%
- Recall : 86.02%
- F1 Score : 86.18%

Classification Report

Class	Precision	Recall	F1-Score	Support
NV	0.94	0.92	0.93	1341
MEL	0.62	0.69	0.65	223
BKL	0.75	0.75	0.75	220
BCC	0.78	0.90	0.84	103
AKIEC	0.73	0.66	0.69	65
VASC	0.92	0.86	0.89	28
DF	0.94	0.70	0.80	23
Accuracy	0.86 (2003 samples)			
Macro Avg	0.81	0.78	0.79	2003
Weighted Avg	0.87	0.86	0.86	2003

TABLE 1 – Classification Report.

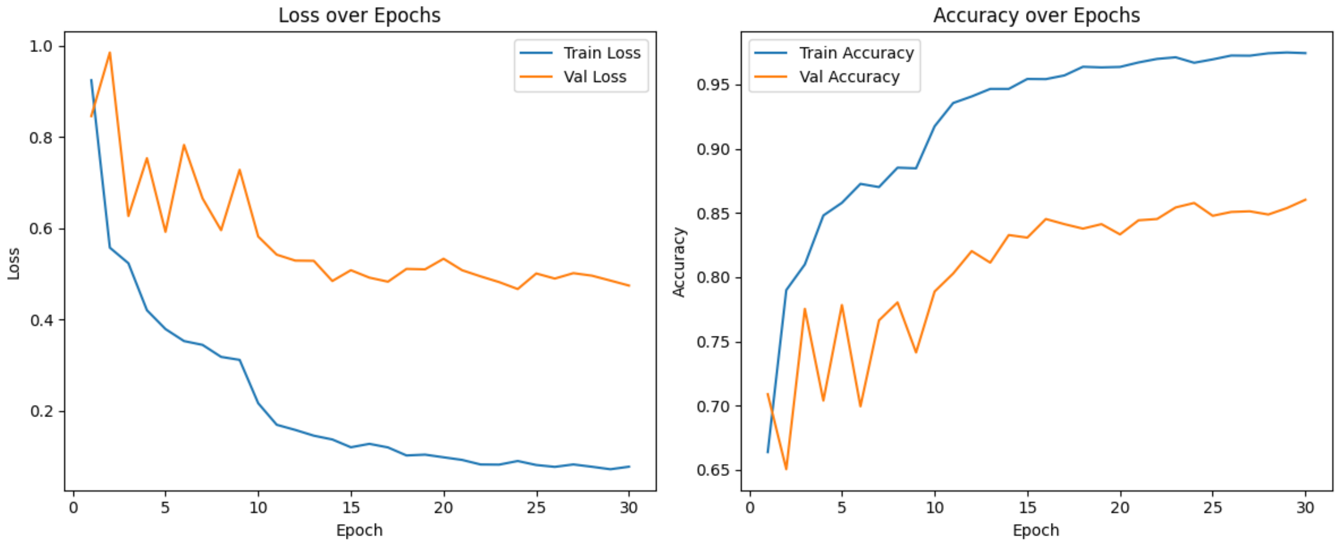

FIGURE 1 – Loss and Accuracy Fonction.

Interpretation

Metric	**Value**	**Interpretation**
Accuracy	86.02%	The model correctly predicts 86% of all cases — a very good performance, especially for medical data.
Precision	86.5%	When the model predicts a class, it is correct most of the time (low false positives).
Recall	86.02%	The model successfully detects most of the true cases (low false negatives).
F1 Score	86.18%	Excellent balance between Precision and Recall — confirms robustness.

Class-Wise Notes

- **NV** : 0.94 (P) / 0.92 (R) / 0.93 (F1) / 1341 Support — Strong detection ; minor confusion with MEL/BKL.
- **MEL** : 0.62 (P) / 0.69 (R) / 0.65 (F1) / 223 Support — 31% of melanomas missed ; improvement opportunity.
- **BKL** : 0.75 (P) / 0.75 (R) / 0.75 (F1) / 220 Support — Balanced performance.
- **BCC** : 0.78 (P) / 0.90 (R) / 0.84 (F1) / 103 Support — High recall — critical for carcinoma detection.
- **AKIEC** : 0.73 (P) / 0.66 (R) / 0.69 (F1) / 65 Support — Some confusion with BKL.

— **VASC** : 0.92 (P) / 0.86 (R) / 0.89 (F1) / 28 Support — Excellent despite few samples.

— **DF** : 0.94 (P) / 0.70 (R) / 0.80 (F1) / 23 Support — Good precision ; some missed cases.

Overall, an accuracy of 86.0%, precision of 86.5%, recall of 86.0%, and F1-score of 86.2% confirms a robust model across all major classes.

0.6 Confusion Matrix Summary

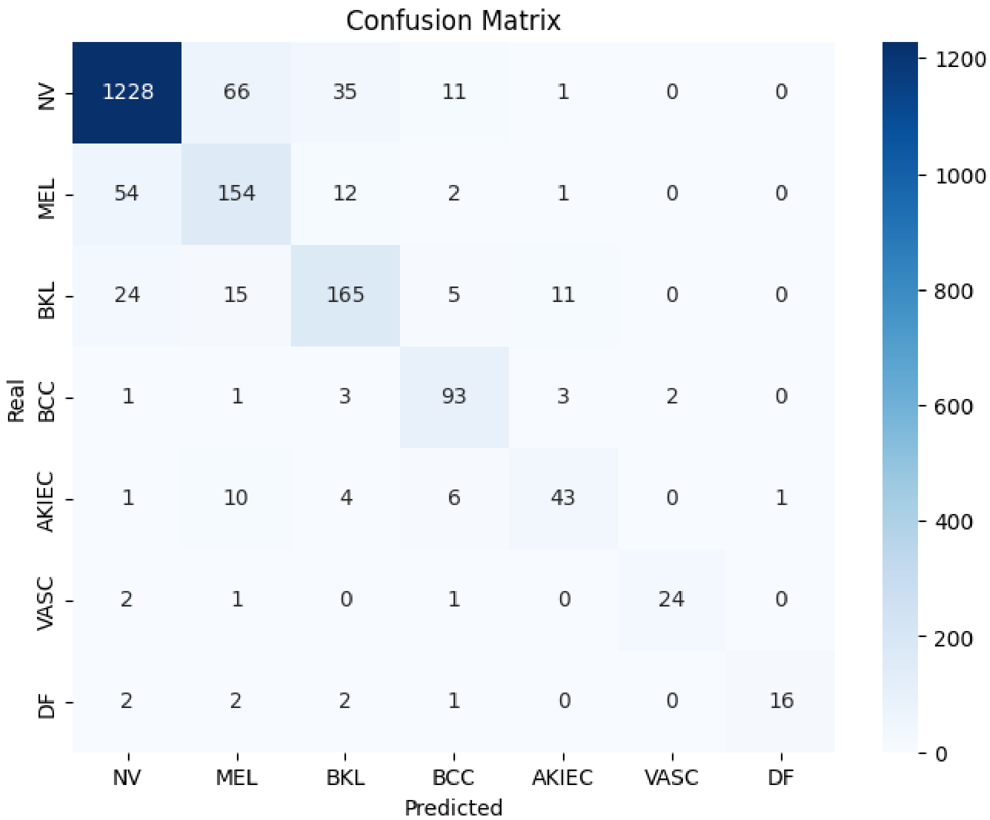

FIGURE 2 – Confusion Matrix.

Real Class	Total Samples	Correct Predictions	Errors	Interpretation
NV (Melanocytic Nevi)	1341	1228 (91.5%)	113	Very well detected. Some confusion with MEL and BKL, which are visually similar.
MEL (Melanoma)	223	154 (69%)	69	Acceptable but about 30% of melanomas are misclassified, which can be critical.
BKL (Benign Keratosis)	220	165 (75%)	55	Good, a few errors confused with MEL and NV.
BCC (Basal Cell Carcinoma)	103	93 (90%)	10	Excellent recall. Critical cancers are detected reliably.
AKIEC (Actinic Keratoses)	65	43 (66%)	22	Some confusion, typical as AKIEC often looks similar to BKL.
VASC (Vascular Lesions)	28	24 (86%)	4	Very well detected.
DF (Dermatofibroma)	23	16 (70%)	7	Good, but some misclassifications happen due to the small sample size.

— Strong diagonal dominance indicates high accuracy.

— Most misclassifications occur between visually similar classes (e.g., NV and MEL).

— Small classes (VASC, DF) were handled well despite fewer samples.

0.7 Advantages & Limitations

Advantages

— **Lightweight & Fast** : 5.3M parameters with high ImageNet performance.

— **Balanced Training** : Weighted sampling and advanced augmentations minimize bias.

— **Efficient Convergence** : Mixed-precision and scheduler accelerate training by 2–3× with stable validation loss.

— **Deployment-Ready** : Optionally quantizable to INT8 for CPU-efficient inference.

Limitations

- **Melanoma Recall** : At 69%, further improvements are needed to reduce false negatives.
- **Static Quantization** : May not fully leverage INT8 on all hardware without QAT.
- **Class Overlap** : Visual similarity (NV vs. MEL, AKIEC vs. BKL) still causes errors.

0.8 Predict Multiple Images

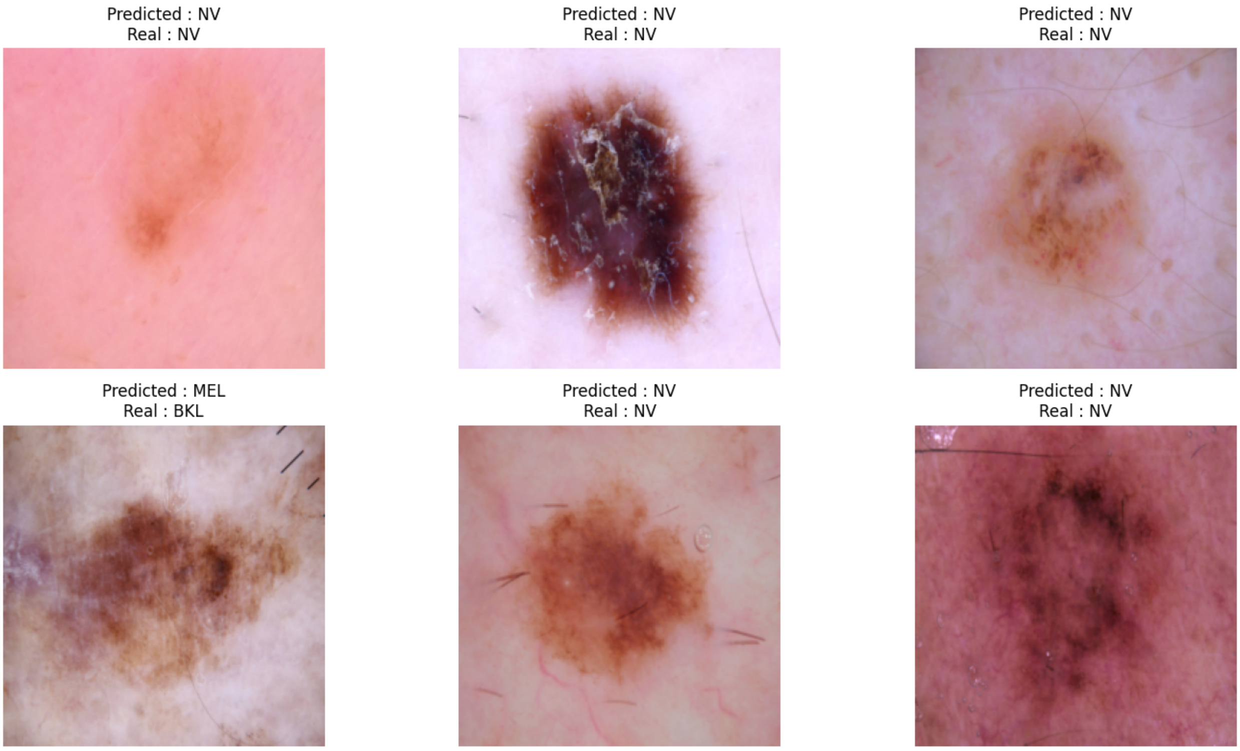

FIGURE 3 – Predict Images.

Conclusion

The developed model achieves high classification performance (86% accuracy) and demonstrates robustness across multiple skin lesion categories. It shows strong potential for clinical decision support systems, helping dermatologists in early and accurate skin cancer detection.

Recommendations

1. Targeted Oversampling or Focal Loss on MEL class to boost sensitivity.

Category	Conclusion
General performance	Excellent (86% accuracy)
Critical errors	Some errors on MELANOMA class (needs cautious deployment)
Robustness	Very good (works well on major and minor classes)
Deployment readiness	Yes, but melanoma detection improvement is recommended.

TABLE 2 – Summury table.

2. Quantization-Aware Training (QAT) to improve INT8 performance across backends.
3. Try Larger EfficientNets (B3/B4) for marginal accuracy gains if compute allows.
4. Augmentation Tuning : Introduce color-spacing or lesion-specific transforms to reduce class confusion.

Recommended Actions

Action	Reason
Oversample or apply Focal Loss for MEL class	To improve melanoma detection and reduce critical errors.
Train for a few more epochs	To slightly improve fine-tuning without overfitting.
Test larger architectures (EfficientNet-B3/B4)	May further boost accuracy at a small compute cost.